

A Multi-hop Reasoning Framework for Cyber Threat Intelligence Knowledge Graph

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Abstract—Cyber threat intelligence is important information for analysing cyber threats. Most of the research focuses on extracting threat entities from threat intelligence and constructing knowledge graphs, with less research on reasoning on the threat intelligence knowledge graph. Existing research has limited ability to reason about implicit information in the threat intelligence knowledge graph and is not interpretable. In addition, the amount of information contained in the threat intelligence knowledge graph has an inherent upper limit, and the existing multi-hop reasoning methods are also constrained by the limitation caused by the knowledge graph, which affects their performance. In this paper, we propose a multi-hop reasoning framework for the cyber threat intelligence knowledge graph, which uses a language model for multi-hop reasoning, views the reasoning process as a sequence-to-sequence task, and generates multi-hop reasoning paths based on reasoning queries. To alleviate the limitation of the threat intelligence knowledge graph, we inject external knowledge graphs into the language model and add a rule enhancement strategy, and the experimental results show some improvement in multi-hop reasoning performance and interpretability.

Index Terms—Cyber Threat Intelligence, Knowledge Graph, Multi-hop Reasoning, Language Model, Interpretability.

I. INTRODUCTION

With the recent dramatic changes in the international geopolitical and economic situation, cyber threat activities have intensified, and the increasing ability of geopolitical effects to influence threat activities in cyberspace¹. Kaspersky² predicts that APT attacks will increase in 2024, and supply chain attacks will create a greater threat, as evidenced by the recent well-known backdoor incident of the xz official repository [1], with such supply chain style attacks lurking close to 3 years, previous reactive cybersecurity defence strategies are hardly effective.

To enhance proactive response to threat activity, Gartner, an IT research and consulting firm, has developed the concept of threat intelligence [2]. Security vendors gather threat information by tracking attack behaviours, capturing malicious samples, monitoring black markets and dark web platforms, and consolidating it into threat intelligence. By sharing and

analyzing this intelligence, it becomes possible to more effectively predict potential cyber threats and develop targeted cyber defence mechanisms.

Current research on reasoning with threat intelligence knowledge graphs mainly utilizes a method of reasoning based on embedding the knowledge graph. The basic fundamental involves embedding entities and relations into the vector space to capture potential semantics. A scoring mechanism is designed to evaluate the alignment of the entities and relations, with the entity having the highest score being considered as the result of reasoning. However, Reasoning methods that rely solely on knowledge graph embedding tend to overlook the symbolic compositionality of relations, which can be ineffective in scenarios where multiple entities and relations collectively influence the reasoning result.

Multi-hop reasoning effectively addresses this issue by synthesizing and analyzing information along multi-hop paths. The process of reasoning is evident and interpretable, making multi-hop reasoning more suitable than previous methods for handling complex and covert situations such as cyber threat activities. As illustrated in Fig. 1, the process of deriving the attack target from the threat entity Silence is uninterpretable, making it difficult to comprehend how the model arrived at its conclusion, whereas in the reasoning process of the threat entity APT32, it can be learned that APT32 attacked China's infrastructure using RemyRAT, and reasoned out that the target of its attack is China. Multi-hop reasoning provides access to the paths taken during the reasoning process, enabling a clear understanding of the final result and facilitating explanations. This capability is crucial for aiding cyber defence decision-making, which often requires consideration of the rationale behind the reasoning results.

Existing multi-hop reasoning methods, although varying in method design, all obtain information from within the knowledge graph data, and their performance is ultimately constrained by the inherent upper limit of the amount of information contained in the knowledge graph data. For example, when it is difficult to reason about geopolitically related cyber-attack information based on existing threat intelligence, it is expected that access to geopolitically related external knowledge will help in the threat intelligence knowledge graph reasoning. In this paper, we propose a multi-hop reasoning

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¹<https://industrialcyber.co/features/growing-convergence-of-geopolitics-and-cyber-warfare-continue-to-threaten-ot-and-ics-environments-in-2024>

²<https://www.kaspersky.com/about/press-releases/ahead-of-the-curve-kasperskys-projections-for-2024s-advanced-threats-landscape>

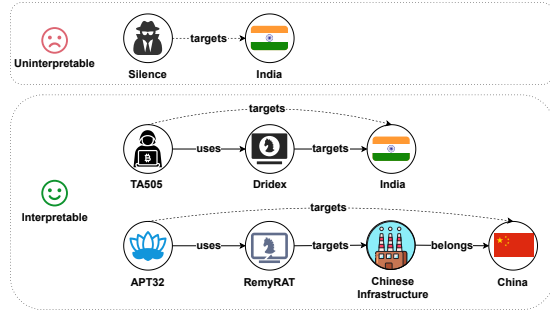


Fig. 1. Illustration of the reasoning process. The top part is the embedding-based reasoning process and the bottom part is the multi-hop reasoning process.

framework for the knowledge graph of cyber threat intelligence, which employs pre-trained language model (PLM) T5 [3] for multi-hop reasoning, injecting external knowledge graphs into the language model to provide relevant background knowledge for threat intelligence reasoning and incorporating rule enhancement strategies as a way of improving the reasonableness of reasoning path generation. Experimental results demonstrate that our method surpasses existing methods in terms of both multi-hop reasoning performance and interpretability.

The paper's contributions can be summarized as follows:

- We propose a multi-hop reasoning framework for the knowledge graph of cyber threat intelligence that introduces a language model for multi-hop reasoning, which is more capable of reasoning about the implicit information of threat entities and is interpretable compared to previous reasoning methods.
- Aiming at the problem that multi-hop reasoning of knowledge graph is limited by the data of the knowledge graph itself, we inject external knowledge graphs into the language model, which further improves the multi-hop reasoning ability.
- To improve the interpretability of multi-hop reasoning, we design a rule enhancement strategy that incorporates logic rules mined from the knowledge graph.

II. RELATED WORKS

In this section, we will review the related work on extracting threat entities and relations, as well as constructing a knowledge graph for threat intelligence. Subsequently, we will review the related research on reasoning with the knowledge graph of cyber threat intelligence and multi-hop reasoning.

A. Threat Intelligence Knowledge Graph

In recent studies focusing on the threat intelligence knowledge graph, [4] developed an open threat intelligence knowledge graph by utilizing an attention-based open information extraction model. [5] established a threat intelligence knowledge graph associated with APT attacks through the extraction of threat entities using a BERT-based named entity recognition

model. [6] employed attack technique templates to gather technical knowledge from threat intelligence and utilized an enhanced algorithm for aligning graphs to detect the attack techniques in the templates, resulting in the creation of the first attack technique knowledge graph. [7] applied a joint model for extracting entity relations from threat intelligence and proposed an enhanced Levenshtein distance algorithm for consolidating entities from various origins into the cohesive threat intelligence knowledge graph.

B. Threat Intelligence Knowledge Graph Reasoning

In comparison to extracting entity relations for threat intelligence, there has been relatively little research on reasoning in threat intelligence knowledge graphs. [8] suggested using a semi-supervised method to build a threat intelligence knowledge graph and utilized two link prediction models, TransH and TuckER, to reason unknown threat information. [9] integrated threat intelligence and security requirements into a knowledge graph called RCTI and introduced an EGNN model based on graph neural networks for link prediction in RCTI, which is used to discover new insights about security requirements and threat intelligence. [10] applied the reasoning in threat intelligence knowledge graphs to derive potential attack patterns by integrating the knowledge graph into cyber attack identification.

C. Knowledge Graph Multi-hop Reasoning

The investigation into multi-hop reasoning for knowledge graphs has attracted considerable interest from the academic community. [11] utilized reinforcement learning to perform multi-hop reasoning and used a single-hop embedding model for estimating the rewards of the target entity, randomly blocking the edges of the agent to enhance its ability to explore different paths. [12] applied meta-learning methods from the field of computer vision to multi-hop reasoning, acquiring meta-information on a high-frequency relation task to accommodate the learning of a smaller number of relations. [13] proposed a framework for multi-hop reasoning using a transformer-based sequence-to-sequence approach, which has the potential to achieve faster convergence compared to methods based on reinforcement learning. [14] used the prediction of the embedding model to dynamically add paths and guide the search direction of the agent, alleviating difficulties in sparse knowledge graph multi-hop reasoning. [15] designed a reward model that takes into account both the accuracy of answers and the reasonableness of paths, thereby improving reinforcement learning models' ability to avoid generating false paths.

III. METHODOLOGY

In this section, we present our proposed framework for multi-hop reasoning in the knowledge graph of cyber threat intelligence, which mainly consists of a threat intelligence extraction part and a multi-hop reasoning part of the threat intelligence knowledge graph, which is used for threat information reasoning to discover potential threat activities and

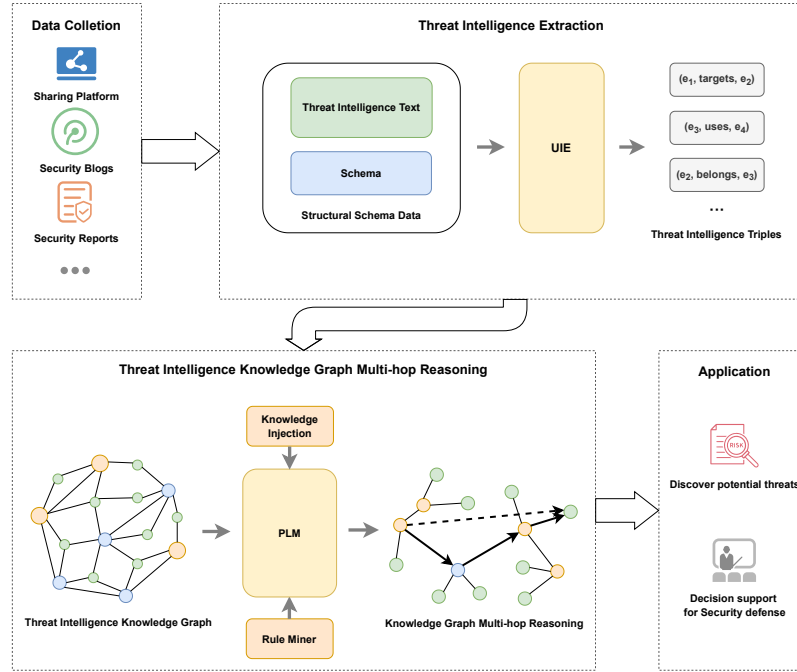


Fig. 2. The Multi-hop Reasoning Framework for Cyber Threat Intelligence Knowledge Graph

threat entities. The framework can be seen in Fig. 2. Within the data collection part, threat intelligence information is gathered from a variety of threat intelligence data sources, and subsequently, in the threat intelligence extraction phase, the Universal Information Extraction (UIE) model [16] is used to extract entity and relation triples from threat intelligence in a joint manner. During the multi-hop reasoning phase, we present a method for conducting multi-hop reasoning on the threat intelligence knowledge graph using a pre-trained language model, which uses the information on the paths and the literal semantics of the entities to reason potential threat information and improves the interpretability of reasoning by injecting external knowledge graphs and incorporating rule enhancement strategies.

A. Data collection

To ensure the effectiveness of threat reasoning, it is essential to collect threat intelligence text data from reliable sources such as threat intelligence sharing platforms, APT reports, security blogs, etc. In this study, we utilized web crawlers and automated scripts to gather a total of 11,459 pieces of threat intelligence text. Subsequently, these data were pre-processed to eliminate irrelevant information in threat intelligence (e.g., report authors or organizations), remove non-ASCII characters contained in the sentences, and employ regular matching techniques to identify key information such as IP addresses, URLs, and domain names. After processing, we obtained a final set of 38,121 sentences for analysis.

Our analysis involved studying the Threat Intelligence corpus and utilizing the STIX standard [17] to categorize entities into 14 categories: “Organization”, “Location”, “Software”, “Malware”, “Indicator”, “Asset”, “Vulnerability”, “Tool”, “Attack pattern”, “Campaign”, “Event”, “Threat Actor”, “Country”, “Group”. Additionally we categorized relations into 11 types: “uses”, “targets”, “indicates”, “exploit”, “launch”, “have”, “belongs”, “exist”, “related to”, “discover”, “alias”. The labelling of threat intelligence sentences was carried out using label-studio [18].

B. Threat intelligence extraction

In the research on extracting entities and relations in threat intelligence, the joint extraction method is superior to the previous pipelined method [7]. We use the UIE model. UIE is a model for generating text-to-structure, which represents different information extraction tasks using a unified format structural extraction language (SEL), using a predefined schema prompt mechanism that guides the model in locating and associating entities and relations.

The first step in extracting threat intelligence using UIE is to define the structured schema instructor (SSI). This schema controls the target range of the UIE model extraction and specifies the entity types and relations to be extracted. The predefined schema in the SSI is shown in Table I. The next step involves accepting the predefined schema instructions and the threat intelligence text as inputs and then selecting the relation extraction subtask. After that, the UIE model encodes the threat intelligence text to generate a hidden representation,

TABLE I
SCHEMA TABLE

Entity	Relation
Organization	uses, targets, exploit, launch, have, belongs, related to, discover, alias
Location	targets, belongs, related to
Software	uses, indicates, belongs, exist, related to
Malware	uses, targets, indicates, exploit, belongs, related to, alias
Indicator	indicates, belongs, related to
Asset	uses, have, belongs, exist, related to
Vulnerability	uses, targets, indicates, exploit, belongs, related to, alias
Tool	uses, indicates, belongs, related to
Attack pattern	uses, targets, indicates, exploit, related to
Campaign	uses, targets, exploit, belongs, related to, alias
Event	indicates, belongs, exist, related to
Threat Actor	uses, targets, exploit, launch, have, belongs, related to, alias
Country	targets, have, belongs, exist, related to
Group	uses, have, belongs, related to

followed by using a decoder to produce a linearized SEL expression. This expression is then converted into relational triples for joint extraction based on SEL expression. Fig. 3 shows part of the threat intelligence knowledge graph.

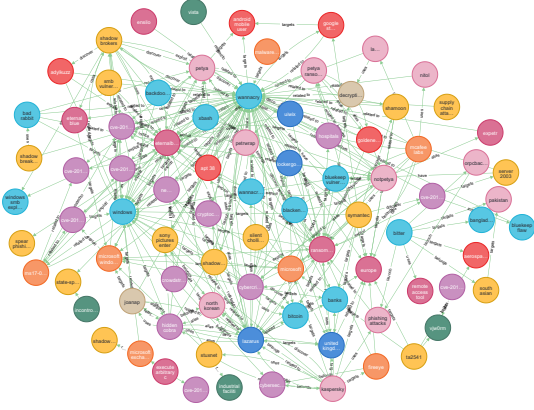


Fig. 3. Part of the Extracted Threat Intelligence Knowledge Graph

C. Threat Intelligence Knowledge Graph Multi-hop Reasoning

After the extraction of the threat intelligence triple as described above, we propose a multi-hop reasoning method for the threat intelligence knowledge graph based on the T5 language model, incorporating a knowledge injection learning process and rule enhancement strategy. During the model's threat training process, the encoder encodes the reasoning query to obtain hidden representations of input sequences. The decoder then generates probability distributions of output sequences based on the encoding results and ground truth embedding. The model's weight is adjusted by calculating loss between output sequences and ground truth using the cross-entropy function.

The model's reasoning process can be seen as a sequence-to-sequence task. The source sequence consists of the threat reasoning query $(h, r, ?)$, while the target sequence includes paths $r_1, t_1, \dots, r_n, t$. This process involves both decoding and encoding, wherein the decoder makes predictions about the current output sequence using the encoding result of the source sequence and previous results of the output sequence until a complete path is reasoned. Specifically, we convert the reasoning query $(h, r, ?)$ and the output path into textual representations suitable for use with the T5 model. The reasoning query is transformed into "predict path: $h - r$ " while the output path is converted to " $r_1 - t_1 - r_2 - t_2 \dots r_n - t$ ". The output path is obtained by searching all paths from h to t and then randomly sampling them. Training data for fine-tuning the pre-trained language model T5 includes both reasoning queries and paths.

In addition, when generating reasoning paths, it is important to constrain the generated content to ensure that the entities and relations on the paths are real. The constraining process is summarized in Algorithm 1. We define a filtering vocabulary based on the threat intelligence triple data and then convert it into a set of token IDs. When the model generates the text, we use this set to filter the generated token sequences, which are finally decoded into an output path.

Algorithm 1 Constrained Processes for Generating Reasoning Paths

Input: Filter vocabulary V
Finetuned model f
input sequence S_{in}
Output: Filtered sequence S_f

- 1: $S_{out} \leftarrow f(S_{in})$ // output sequence
- 2: $S_f = ()$
- 3: $id_V = ()$
- 4: **for** $vocab \in V$ **do**
- 5: // convert the vocabulary V into a set of token ids
- 6: $id_V \leftarrow (id_V, \text{convert}(vocab))$
- 7: **end for**
- 8: **for** $t \in S_{out}$ **do**
- 9: **if** $t \in id_V$ **then**
- 10: $S_f \leftarrow (S_f, t)$
- 11: **end if**
- 12: **end for**
- 13: **return** S_f

Compared to other textual information available on the web, the nature of threat intelligence itself leads to a relative scarcity of threat intelligence data. This is further compounded by limitations in the amount of information within the threat intelligence knowledge graph, which restricts multi-hop reasoning from threat intelligence. On the other hand, knowledge injection can enhance threat intelligence reasoning by adding additional knowledge beyond what is covered in its original training data [19]. [20] and [21] discovered that the FFN layer of the transformer architecture comprises two linear layers capable of generating keys and values. This key-value pair

structure enables the storage of knowledge. We have applied this theory to knowledge injection.

When cybersecurity researchers analyze and reason about threat intelligence, it is important to study the background knowledge associated with it. We incorporate background knowledge related to cyber threat activities into the language model to aid in the reasoning process. This involves converting the external knowledge graph, icews [22], into textual form for link prediction input and output. For example, a triple (h, r, t) would be transformed into input and output text of the form “predict tail: $h - r$ ” and “ t ”. Afterwards, the knowledge injection structure should be incorporated into the FFN layer of the decoder in the language model. We utilize two matrices W'_1 and W'_2 as new key-value pairs, which are appended to the parameter matrices of the original FFN’s linear layer following W_1 and W_2 . This expansion serves to increase the dimensionality of the intermediate hidden states within the FFN layer. The addition process is as follows:

$$FFN'(H) = GELU(HW_1^T)W_2 + GELU(HW_1'^T)W_2'. \quad (1)$$

Where H represents the input of the FFN layer, and $FFN'(H)$ represents the output of the FFN layer following the addition of the matrix. In the process of injecting knowledge into the T5 language model, it is essential to freeze the original parameters of the model and only update the parameters of the newly added knowledge injection structure. This method ensures that the original reasoning performance of the language model remains unaffected while effectively utilizing injected knowledge.

Multi-hop reasoning involves the deduction of indirect conclusions from a set of facts or knowledge through multiple reasoning steps. Rules play a crucial role in providing an explanation for the reasoning process, thus reducing uncertainty and errors. When generating the content, these rules are applied to ensure that the output aligns with the expected criteria and logic. The process of mining rules includes extracting Horn rules from the knowledge graph and iteratively calculating their confidence level based on logical relations between entities in the graph.

To enhance the interpretability of the multi-hop reasoning model, we have incorporated a rule enhancement strategy during the decoding of generated paths. Prior to this process, it is necessary to mine the rules in the threat intelligence knowledge graph. For this purpose, we utilize AMIE+ [23] for rule mining. The format of the mined rules is represented as $r(X, Y) \leftarrow r(X, A_1) \wedge r(A_1, A_2) \wedge \dots \wedge r(A_n, Y)$ where $r(X, Y)$ is equivalent to the knowledge graph triple (X, r, Y) . Each mined rule is associated with a confidence score S_r provided by the rule miner. Additionally, our multi-hop reasoning model generates a decoding score S_d for each sequence of candidate paths. The final path sequence score S indicates its likelihood as the final path of reasoning. The calculation process is summarized in Algorithm 2.

Algorithm 2 Rules Enhancement Strategy

Input: Candidate path sequence set $Path$
mined rules set R_r
constant factor m

Output: Path Sequence Score S

```

1:  $S = ()$ 
2: for  $p \in Path$  do
3:   // The decoding score corresponding to path  $p$ 
4:    $S_d \leftarrow score(p)$ 
5:   for  $r \in R_r$  do
6:     // The confidence score corresponding to rule  $r$ 
7:      $S_r \leftarrow score(r)$ 
8:     if  $match(p, r) = True$  then
9:        $S \leftarrow mS_d + (1 - m)S_r$ 
10:    else
11:       $S \leftarrow (1 - m)S_d$ 
12:    end if
13:  end for
14: end for
15: return  $S$ 

```

IV. EXPERIMENTS

A. Experiment dataset

Threat intelligence data is sourced from threat intelligence sharing platforms, open source articles and blogs within the security community. A series of processes are then conducted on the threat intelligence text to generate 38,121 sentences. Additionally, 5,000 sentences were manually annotated using the label-studio annotation platform. These annotated data were utilized to train the UIE model, which was subsequently employed to extract results from all sentences conforming to the schema format. Following this extraction, the results were imported into label-studio for manual validation, during which incorrectly extracted sentences were removed. This process resulted in obtaining 25,459 threat intelligence knowledge graph triples. To evaluate the performance of our multi-hop reasoning method, we conducted experiments not only on the threat intelligence knowledge graph CSTI that we constructed but also on the Gdelt [24] knowledge graph, which records major global events and is more similar in nature to threat intelligence data. The knowledge graph data is divided into a training set, validation set, and test set in an 8:1:1 ratio. The dataset statistics are shown in Table II. T/E represents the ratio between the number of training sets and the number of entities, which is used to measure the sparseness of the knowledge graph. It can be observed that CSTI is more sparse compared with Gdelt.

TABLE II
SUMMARY STATISTICS FOR DATASETS

Dataset	Entity	Relation	Train	Dev	Test	T/E
Gdelt	7,691	240	103,844	12,980	12,981	13.5
CSTI	6,386	11	20,367	2,546	2,546	3.18

TABLE III
COMPARISON OF RESULTS FROM MULTI-HOP REASONING EXPERIMENTS

Dataset	CSTI				Gdelt			
	<i>MRR</i>	<i>Hits@1</i>	<i>Hits@3</i>	<i>Hits@10</i>	<i>MRR</i>	<i>Hits@1</i>	<i>Hits@3</i>	<i>Hits@10</i>
MultiHopKG	0.204	0.141	0.222	0.334	0.271	0.224	0.291	0.358
Meta-KGR	0.180	0.122	0.191	0.294	0.267	0.215	0.288	0.366
DacKGR	0.176	0.114	0.190	0.311	0.260	0.217	0.278	0.334
SQUIRE	0.203	0.135	0.226	0.341	0.288	0.230	0.311	0.400
PSAgent	0.206	0.140	0.225	0.330	0.268	0.215	0.288	0.364
Ours	0.215	0.166	0.250	0.359	0.316	0.292	0.385	0.469

B. Evaluation metrics

In previous experiments on multi-hop reasoning, the confidence of each candidate path was calculated as the ranking score for evaluation. We decoded a fixed number of candidate path sequences to maintain consistent evaluation criteria and used the logarithm of the decoding scores as the ranking score for evaluation. For knowledge graph reasoning performance experiments, two commonly used metrics, *MRR* and *Hits@K*, are employed. Following this design, we compute these two metrics as follows:

- 1) *Hits@K* represents the proportion of correct tail entities among the top K predicted entities in the ranking ($K = 1, 3, 10$):

$$Hits@K = \frac{|\{q \in Q : rank_q \leq K\}|}{|Q|}. \quad (2)$$

- 2) *MRR* is the average inverse rank of all correct tail entities:

$$MRR = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{rank_q}. \quad (3)$$

where q is the correct tail entity, Q is the set of all tail entities predicted, and $rank_q$ is the ranking of q .

C. Experiment settings

To evaluate the efficiency of our proposed multi-hop reasoning approach, we conducted a comparison with previous multi-hop reasoning models, including MultiHopKG [11], Meta-KGR [12], DacKGR [14], SQUIRE [13], and PSAGENT [15]. We performed experiments on CSTI and Gdelt datasets using a T5 language model with a vector dimension size set to 512 dimensions and an FFN layer dimension set to 2048. For CSTI, we utilized the adafactor adaptive optimizer for the language model with a batch size of 64 and implemented a dropout mechanism with a value of 0.1 to prevent overfitting. The number of epochs was set to 80. For Gdelt, the batch size was set to 128 and epochs were set to 120. All experiments were carried out on Tesla T4 GPUs and checkpoints with the best validation performance were chosen for testing purposes.

D. Experiment results

In Table III, we present a comparison of the performance of our multihop reasoning method with other multihop reasoning models on CSTI and Gdelt. On CSTI, our method achieves

an *MRR* metric of 21.5, representing a 4.3% improvement compared to the previous model with the best performance. On Gdelt, Our method achieves an *MRR* metric of 31.6, representing a 9.7% improvement over the previous model with the best performance. Additionally, all other test metrics also show partial improvements in the experiments.

The results from CSTI and Gdelt demonstrate that our method effectively leverages the information within the knowledge graph for multi-hop reasoning. The superior performance on Gdelt indicates that the abundance of information in the densely populated knowledge graph is beneficial for reasoning, highlighting the importance of incorporating external knowledge for threat intelligence reasoning. Despite the sparsely structured nature of knowledge graphs like CSTI, our method still achieves optimal results thanks to the inherent knowledge within the language model and the additional injected knowledge.

E. Ablation experiment

To investigate the impact of incorporating external knowledge graphs and rule enhancement strategies on reasoning performance, we conducted ablation experiments. Specifically, we performed three ablation experiments on CSTI and Gdelt, where we removed the injections of external knowledge graphs (-k) and the application of rule enhancement strategies (-r). The results of these experiments are presented in Table IV, indicating that both methods contribute to improved multi-hop reasoning performance. Furthermore, it is evident that the injection of external knowledge graphs has a more significant impact on multi-hop reasoning compared to the modest improvement achieved by the rule enhancement strategy.

F. Interpretability experiment

Multi-hop reasoning offers reasoning paths to explain the results of threat information reasoning. The accuracy of the reasoning results is undeniably crucial, as well as the reasonability and interpretability of the reasoned paths. To evaluate the interpretability of our proposed multi-hop reasoning method using quantitative measures, we adopt Lv's framework [25], which includes three evaluation metrics PR, LI, and GI for assessing the interpretability of multi-hop reasoning models:

- 1) PR represents the percentage of test set triples where the model successfully identifies a path from the head entity

TABLE IV
COMPARATIVE RESULTS OF ABLATION EXPERIMENTS

Dataset	CSTI				Gdelt			
	<i>MRR</i>	<i>Hits@1</i>	<i>Hits@3</i>	<i>Hits@10</i>	<i>MRR</i>	<i>Hits@1</i>	<i>Hits@3</i>	<i>Hits@10</i>
Ours -k -r	0.209	0.164	0.241	0.358	0.307	0.277	0.369	0.457
Ours -k	0.209	0.161	0.248	0.361	0.309	0.282	0.375	0.463
Ours -r	0.211	0.166	0.243	0.357	0.311	0.284	0.380	0.464
Ours	0.215	0.166	0.250	0.359	0.316	0.292	0.385	0.469

to the tail entity. This metric is utilized to evaluate the model's proficiency in path discovery.

$$PR = \frac{\sum_{tr \in Q_{test}} Cnt(tr)}{|Q_{test}|}. \quad (4)$$

Where tr represents the triple (h, r, t) , Q_{test} represents the test set, and $Cnt(tr)$ is the indicator function that determines whether the path can be found.

- 2) LI aims to evaluate the reasonableness of the paths identified by the model, specifically whether the explanations provided by the model at a local level align with the actual or anticipated logical relations.

$$LI = \frac{\sum_{tr \in Q_{test}} Cnt(tr)S(p)}{\sum_{tr \in Q_{test}} Cnt(tr)}. \quad (5)$$

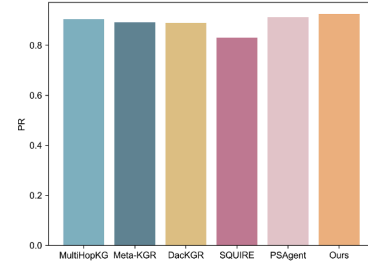
Where $S(p)$ represents the interpretability score of path p , which is generated using the automated mining method AnyBURL [26].

- 3) GI is an evaluation of the model's overall interpretability.

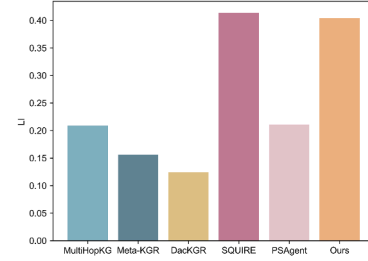
$$GI = PR \cdot LI. \quad (6)$$

We conducted experiments to compare the interpretability of previous multi-hop reasoning models with our proposed method. The results are presented in Fig. 4, showing the performance on CSTI. It is evident that the previous reasoning models exhibit relatively poor performance in the metric of LI, with the exception of SQUIRE, leading to overall poor interpretability. Our method achieves optimality in the index of PR due to the constraint mechanism. However, there exists a certain gap between our method and the optimal model in terms of LI, indicating a lack of local interpretability in the multi-hop reasoning method based on pre-trained language models. Our method still has been optimized for overall interpretability.

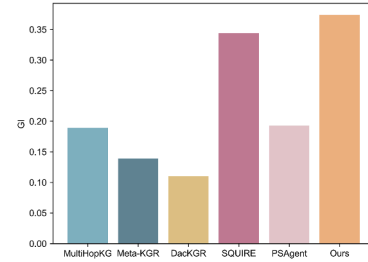
In order to demonstrate the effectiveness of our framework applied in threat intelligence knowledge graph reasoning, we present three case studies in Table V. These case studies include attack attribution, attack target discovery, and threat activity correlation. Attack attribution involves a step-by-step reasoning of the attacker through the attacked victim. Attack target discovery entails the reasoning of the attacker's next target to be attacked based on the attacking organization. Lastly, threat activity correlation involves correlating the attack activity with other attack activities or attackers based on current monitoring. In these cases, the relation is followed by



(a) PR results



(b) LI results



(c) GI results

Fig. 4. Comparative results of interpretability experiments

“rev” to indicate an inverse relation. The Hits@1 reasoning path is provided in the case study.

V. CONCLUSION

This paper proposes a multi-hop reasoning framework for the cyber threat intelligence knowledge graph. We utilize a unified information extraction model to extract threat intelligence text, which is then constructed into a preliminary threat intelligence knowledge graph. We employ a pre-trained

TABLE V
CASE STUDY

Attack Attribution	Reasoning Query: (<i>BBC, targets rev, ?</i>) Path: BBC $\xrightarrow{uses}$ MOVEit $\xrightarrow{exist}$ CVE-2023-34362 $\xrightarrow{exploit rev}$ Cl0p ransomware organization
Attack Target Discovery	Reasoning Query: (<i>APT32, targets, ?</i>) Path: APT32 $\xrightarrow{uses}$ COVID-19 phishing documents $\xrightarrow{exist}$ DenesRAT $\xrightarrow{targets}$ Chinese medical institutions
Threat Activity Correlation	Reasoning Query: (<i>Q NAP device attack activity, launch rev, ?</i>) Path: Q NAP device attack activity $\xrightarrow{related to}$ Q NAP devices $\xrightarrow{targets rev}$ Raspberry Robin $\xrightarrow{uses rev}$ Evil Corp

language model for multi-hop reasoning, integrating an external knowledge graph and incorporating rule enhancement strategies. Our method leverages information on the reasoning path and demonstrates superior capability in reasoning about threat information compared to previous multi-hop reasoning methods. Moreover, our method offers enhanced interpretability as compared to prior work on threat intelligence knowledge graph reasoning. This significantly aids in understanding and dissecting complex threat activities. Regarding future work, we plan to improve the extraction module by incorporating multi-layered information such as vulnerabilities, assets, and threats into the threat intelligence knowledge graph for reasoning.

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REFERENCES

- [1] Redhat, "Cve-2024-3094," 2024. [Online]. Available: <https://access.redhat.com/security/cve/CVE-2024-3094>
- [2] R. McMillan, "How gartner defines threat intelligence," 2016. [Online]. Available: <https://www.gartner.com/en/documents/3222217>
- [3] C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu, "Exploring the limits of transfer learning with a unified text-to-text transformer," *Journal of machine learning research*, vol. 21, no. 140, pp. 1–67, 2020.
- [4] I. Sarhan and M. Spruit, "Open-cykg: An open cyber threat intelligence knowledge graph," *Knowledge-based systems*, vol. 233, p. 107524, 2021.
- [5] Y. Ren, Y. Xiao, Y. Zhou, Z. Zhang, and Z. Tian, "Cskg4apt: A cybersecurity knowledge graph for advanced persistent threat organization attribution," *IEEE Transactions on Knowledge and Data Engineering*, 2022.
- [6] Z. Li, J. Zeng, Y. Chen, and Z. Liang, "Attack: Constructing technique knowledge graph from cyber threat intelligence reports," in *European Symposium on Research in Computer Security*. Springer, 2022, pp. 589–609.
- [7] Y. Guo, Z. Liu, C. Huang, N. Wang, H. Min, W. Guo, and J. Liu, "A framework for threat intelligence extraction and fusion," *Computers & Security*, vol. 132, p. 103371, 2023.
- [8] N. Rastogi, S. Dutta, A. Gittens, M. J. Zaki, and C. Aggarwal, "Tinker: A framework for open source cyberthreat intelligence," in *2022 IEEE International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)*. IEEE, 2022, pp. 1569–1574.
- [9] Y. Zhang, J. Chen, Z. Cheng, X. Shen, J. Qin, Y. Han, and Y. Lu, "Edge propagation for link prediction in requirement-cyber threat intelligence knowledge graph," *Information Sciences*, vol. 653, p. 119770, 2024.
- [10] E. Gilliard, J. Liu, and A. A. Aliyu, "Knowledge graph reasoning for cyber attack detection," *IET Communications*, 2024.
- [11] X. V. Lin, R. Socher, and C. Xiong, "Multi-hop knowledge graph reasoning with reward shaping," in *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, 2018, pp. 3243–3253.
- [12] X. Lv, Y. Gu, X. Han, L. Hou, J. Li, and Z. Liu, "Adapting meta knowledge graph information for multi-hop reasoning over few-shot relations," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 2019, pp. 3376–3381.
- [13] Y. Bai, X. Lv, J. Li, L. Hou, Y. Qu, Z. Dai, and F. Xiong, "Squire: A sequence-to-sequence framework for multi-hop knowledge graph reasoning," in *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, 2022, pp. 1649–1662.
- [14] X. Lv, X. Han, L. Hou, J. Li, Z. Liu, W. Zhang, Y. Zhang, H. Kong, and S. Wu, "Dynamic anticipation and completion for multi-hop reasoning over sparse knowledge graph," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 5694–5703.
- [15] C. Jiang, T. Zhu, H. Zhou, C. Liu, T. Deng, C. Hu, and J. Li, "Path spuriousness-aware reinforcement learning for multi-hop knowledge graph reasoning," in *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, 2023, pp. 3181–3192.
- [16] Y. Lu, Q. Liu, D. Dai, X. Xiao, H. Lin, X. Han, L. Sun, and H. Wu, "Unified structure generation for universal information extraction," in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2022, pp. 5755–5772.
- [17] STIX Standard, OASISOPEN. Std., Rev. 2.1, 2021. [Online]. Available: <https://docs.oasis-open.org/cti/stix/v2.1/os/stix-v2.1-os.html>
- [18] HumanSignal, "label-studio," 2024. [Online]. Available: <https://labels.tud.io/>
- [19] W. Liu, P. Zhou, Z. Zhao, Z. Wang, Q. Ju, H. Deng, and P. Wang, "K-bert: Enabling language representation with knowledge graph," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 34, no. 03, 2020, pp. 2901–2908.
- [20] D. Dai, L. Dong, Y. Hao, Z. Sui, B. Chang, and F. Wei, "Knowledge neurons in pretrained transformers," in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2022, pp. 8493–8502.
- [21] Q. Dong, D. Dai, Y. Song, J. Xu, Z. Sui, and L. Li, "Calibrating factual knowledge in pretrained language models," in *Findings of the Association for Computational Linguistics: EMNLP 2022*, 2022, pp. 5937–5947.
- [22] Z. Wang, H. Du, Q. Yao, and X. Li, "Search to pass messages for temporal knowledge graph completion," in *Findings of the Association for Computational Linguistics: EMNLP 2022*, 2022, pp. 6160–6172.
- [23] L. Galárraga, C. Teflioudi, K. Hose, and F. M. Suchanek, "Fast rule mining in ontological knowledge bases with amie+," *The VLDB Journal*, vol. 24, no. 6, pp. 707–730, 2015.
- [24] W. Jin, M. Qu, X. Jin, and X. Ren, "Recurrent event network: Autoregressive structure inference over temporal knowledge graphs," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 6669–6683.
- [25] X. Lv, Y. Cao, L. Hou, J. Li, Z. Liu, Y. Zhang, and Z. Dai, "Is multi-hop reasoning really explainable? towards benchmarking reasoning interpretability," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 2021, pp. 8899–8911.
- [26] C. Meilicke, M. W. Chekol, D. Ruffinelli, and H. Stuckenschmidt, "Anytime bottom-up rule learning for knowledge graph completion," in *Proceedings of the 28th International Joint Conference on Artificial Intelligence*, 2019, pp. 3137–3143.